The Primitive Measure Theorem: Constructive Measure and Minimal Simplex Closure

Joseph Fredrickson
Independent Researcher
Orem, Utah, USA
josephfredrickson@live.com
ORCID: 0009-0005-3327-4408

May 21, 2026
Revised June 11, 2026

Abstract

This paper formalizes a distinction between coordinate representation and constructive primitive geometry. Points, lines, planes, squares, cubes, and limits are powerful tools for representation and calculation, but a positive measure is not constructively obtained from zero-measure primitives by ordinary finite or countable aggregation. If a primitive is required to consume space rather than merely locate, parameterize, or represent it, then it must already possess the minimal dimensional closure required by the measure it contributes. For finite vertex-defined closed primitives, the minimal positive-measure primitive in dimension n is the n-simplex. Thus, in two dimensions the minimal area-consuming primitive is the triangle; under maximal balance symmetry, the equilateral triangle. In three dimensions the minimal volume-consuming primitive is the tetrahedron; under maximal balance symmetry, the regular tetrahedron. This result does not reject calculus, measure theory, or Cartesian coordinates. It clarifies the cost of mistaking an axiomatic continuum or coordinate scaffold for a constructive ontology of consumed space.

Keywords: constructive geometry; primitive measure; simplex closure; foundations of geometry; measure theory; philosophy of mathematics

1 Introduction

Modern mathematics and physics frequently represent objects using points and coordinate systems, and standard treatments of analysis, measure, and geometry provide powerful formal frameworks for doing so (Rudin, 1976; Federer, 1969; Coxeter, 1973). These representations are indispensable, but they are sometimes treated as if they were constructively primitive. A point has no length, area, or volume. A line has length but no area or volume. An ideal plane may represent area, but it has no depth and therefore no volume. A square or cube may serve as a convenient coordinate unit, but neither is the minimal closed primitive of its dimension.

The issue is not whether these tools are valid, nor whether constructive mathematics already provides alternative foundations for analysis and proof (Bishop, 1967). They are valid within their proper domains. The issue is whether a zero-measure object can serve as the constructive primitive of positive measure in the sense of consumed space. It cannot do so by ordinary finite or countable

aggregation:

$$\sum_{i=1}^{m} x_i \cdot 0 = 0. \tag{1}$$

This equation is elementary algebra, but it becomes foundational when a model treats zero-measure locators as if they were the constructive source of positive measure. If every alleged primitive contribution has literal zero measure, then finite or countable accumulation of those contributions cannot yield positive measure. A framework that obtains positive measure from zero-measure points is therefore using something beyond ordinary constructive addition of zero-measure parts: an axiomatically completed continuum, an uncountable whole, a limiting process, an extension of the primitive domain, or some other operation whose finite stages are not literal zero-width parts.

This paper proves a conditional theorem. If positive measure is interpreted constructively, and if the primitive is required to consume the relevant dimension of space as a finite vertex-defined closed primitive, then the primitive must be the minimum closed dimensional form: a simplex. In two dimensions this is a triangle; under balance, an equilateral triangle. In three dimensions this is a tetrahedron; under balance, a regular tetrahedron.

2 Scope and Non-Claims

Because the argument is simple, it is easy to mischaracterize. The following clarifications are essential.

NC1. This paper does not claim that calculus is invalid.

NC2. This paper does not claim that measure theory is invalid.

NC3. This paper does not claim that Cartesian coordinates are useless or false as measurement tools.

NC4. This paper does not claim that a continuum cannot be defined axiomatically.

NC5. This paper does not claim that circles, spheres, squares, or cubes are unusable mathematical objects.

The claim is narrower: positive measure is not constructively obtained from literal zero-measure primitives by ordinary finite or countable aggregation. Therefore, if one demands a constructive primitive that itself contributes positive measure, that primitive must already possess nonzero dimensional closure.

3 Primitive Counting and Constructive Contribution

The constructive account of measure also requires a primitive account of number. In this paper, number is not treated first as an abstract symbol, binary encoding, or coordinate label. A number is treated constructively as a count of distinguishable objects, states, events, or process completions.

Thus the primitive numerical act is iterative marking:

$$1, \quad 1+1, \quad 1+1+1, \quad \ldots \tag{2}$$

Each increment corresponds to an additional distinguishable completion. Binary notation, decimal notation, algebraic notation, and coordinate labels are secondary encodings of this prior counting act. They are not the primitive source of number.

This distinction matters because constructive measure presupposes countable contribution. A constructive measure cannot begin with a zero-measure contribution and then obtain positive measure merely by relabeling, encoding, or limiting that contribution. The mark must correspond to a nonzero distinguishable contribution in the relevant dimension of measure.

In this sense, zero is not a constructive part. Zero denotes absence of contribution. A model that treats literal zero-measure elements as the constructive source of positive measure confuses absence with primitive contribution.

The primitive counting rule may be stated as follows.

> A constructive number counts completed distinguishable contributions. A notation may encode the count, but the notation does not create the counted object, state, event, or process.

This rule prevents circularity. One may not define number by an encoding whose meaning already presupposes counting. One may not define measure by a limit whose meaning already presupposes nonzero finite contributions. Likewise, one may not define physical progression by a coordinate axis whose meaning already presupposes ordered state change.

Constructive arithmetic therefore begins with iterative completion, not with symbolic abstraction. Addition counts successive completions. Multiplication counts repeated groups of completions. Exponents count repeated dimensional or operational composition. These operations are valid, but their constructive meaning derives from prior iterative counting.

4 Coordinate Measure and Constructive Measure

Definition 4.1 (Coordinate measure)**.** *A coordinate measure assigns magnitude to a region using a chosen coordinate grammar, such as orthogonal Cartesian axes. Within such a grammar, area and volume are commonly expressed by products such as*

$$A = xy, \qquad V = xyz. \tag{3}$$

These formulas describe measurement after a coordinate framework has been imposed. They do not prove that squares or cubes are constructively primitive.

Definition 4.2 (Constructive measure)**.** *A constructive measure concerns the kind of primitive object that can contribute positive measure in the first place. A constructive primitive is not merely a locator or parameter; it must consume the relevant dimensional measure.*

Definition 4.3 (Consumed space)**.** *Let μ_n denote an n-dimensional measure. A primitive P consumes n-dimensional space if*

$$\mu_n(P) > 0. \tag{4}$$

It fails to consume n-dimensional space if

$$\mu_n(P) = 0. \tag{5}$$

Definition 4.4 (Dimensional closure)**.** *A primitive has dimensional closure in dimension n if it bounds a finite n-dimensional region. Closure is necessary for consumed space: without closure, one has location, extension, boundary, or orientation, but not an enclosed positive n-dimensional region.*

Definition 4.5 (Finite vertex-defined primitive)**.** *A finite vertex-defined primitive in dimension n is a closed finite cell determined by a finite set of vertices and their boundary relations. Its positive measure is evaluated as n-dimensional measure of the bounded region it encloses.*

The main theorem of this paper concerns finite vertex-defined constructive primitives. It does not assert that every positive-measure set in measure theory is a simplex. It asserts that the minimum finite closed vertex-defined object capable of consuming positive n-dimensional measure is the n-simplex.

Table 1 summarizes the distinction used throughout the paper. The table is not a replacement for the formal argument; it is a guide to the roles played by common geometric objects when they are interpreted as coordinate tools rather than as constructive primitives.

Table 1: Coordinate role and constructive primitive status of common objects

Object	**Coordinate or formal role**	**Constructive primitive status**
Point	Locator or coordinate label	Zero measure; not a length, area, or volume primitive
Line segment	One-dimensional extension relation	Length primitive; not an area or volume primitive
Triangle	Closed two-dimensional boundary	Minimal finite closed area-consuming primitive
Square or rectangle	Coordinate area unit or reporting frame	Composite relative to triangular closure
Tetrahedron	Closed three-dimensional boundary	Minimal finite closed volume-consuming primitive
Cube	Coordinate volume unit or reporting frame	Composite relative to tetrahedral closure
Limit	Refinement or completion tool	Constructively valid only when finite-stage primitive structure is preserved or an extension is declared

5 The Zero-Contribution Lemmas

Lemma 5.1 (Finite zero-measure aggregation)**.** *Let $P_1, \ldots, P_m$ be primitives satisfying $\mu(P_i) = 0$ for all i. Assuming finite subadditivity and nonnegative measure,*

$$\mu\left(\bigcup_{i=1}^{m} P_i\right) = 0. \tag{6}$$

Proof. By finite subadditivity,

$$\mu\left(\bigcup_{i=1}^{m} P_i\right) \leq \sum_{i=1}^{m} \mu(P_i) = \sum_{i=1}^{m} 0 = 0. \tag{7}$$

Since measure is nonnegative, the measure of the union is exactly zero. □

Lemma 5.2 (Countable zero-measure aggregation)**.** *Let $\{P_i\}_{i=1}^{\infty}$ be a countable collection of primitives satisfying $\mu(P_i) = 0$ for all i. Assuming countable subadditivity and nonnegative measure,*

$$\mu\left(\bigcup_{i=1}^{\infty} P_i\right) = 0. \tag{8}$$

Proof. By countable subadditivity,

$$\mu\left(\bigcup_{i=1}^{\infty} P_i\right) \leq \sum_{i=1}^{\infty} \mu(P_i) = \sum_{i=1}^{\infty} 0 = 0. \tag{9}$$

Again, by nonnegativity, the measure is exactly zero. □

Corollary 5.3 (No positive measure from zero contributions)**.** *Positive measure cannot be constructively produced by ordinary finite or countable aggregation of zero-measure primitives.*

Proof. The result follows immediately from the preceding lemmas. □

Remark 5.4. *This result does not contradict standard measure theory. A continuum may be assigned positive measure even though each point has measure zero, because the continuum is not constructed as an ordinary finite or countable sum of point measures. It is treated as an axiomatically completed object. The distinction is between coordinate or axiomatic representation and constructive aggregation from primitive contributions.*

6 The Limit-Cost Principle

Principle 6.1 (Limit-cost principle)**.** *A limit operation may be mathematically valid while still destroying the structure required for constructive interpretation. If a primitive is claimed to construct positive n-dimensional measure, then any limiting operation applied to that primitive must preserve the minimum structure required to contribute nonzero n-dimensional measure. If the limit removes that structure, then the resulting object is no longer a constructive primitive of the original measure.*

Remark 6.2. *A triangle whose height is sent to zero becomes degenerate and has zero area. A tetrahedron whose height or depth is sent to zero becomes degenerate and has zero volume. Therefore, one may assert a limit, but one must also account for the cost of what the limit destroys.*

7 The Nonzero-Depth Admission

The constructive use of calculus requires an often-unstated admission: before a limit is taken, the relevant subdivision must remain nonzero in the dimension being measured. If the subdivision is replaced by literal zero before the operation is evaluated, the constructive quantity collapses.

For example, a Riemann-type contribution has the form

$$f(x_i^*)\Delta x_i. \tag{10}$$

If Δx_i is substituted with literal zero at the constructive stage, then

$$f(x_i^*) \cdot 0 = 0, \tag{11}$$

and the finite sum contributes no area:

$$\sum_i f(x_i^*) \cdot 0 = 0. \tag{12}$$

Thus the calculus operation retains constructive content because Δx_i remains nonzero throughout each finite approximation and is sent toward a limiting condition only after the nonzero contributions have been formed.

The same issue appears in differentiation. The difference quotient

$$\frac{f(x+h)-f(x)}{h} \tag{13}$$

requires $h \neq 0$ before the limiting operation is taken. Substituting $h = 0$ directly gives an undefined quotient. Therefore, the derivative is not obtained by evaluating change over literal zero extension. It is obtained by preserving a nonzero interval until the limiting relation is formed.

This gives the nonzero-depth admission.

> A constructive limit may approach zero within a formalism, but the constructive operation itself requires nonzero dimensional content before the limiting abstraction is taken. Literal zero cannot carry the measure, relation, or causal constraint being modeled.

8 Primitive-Bound Limits and Constructive Curves

The Limit-Cost Principle implies that constructive limits cannot be interpreted as limits to literal zero whenever the object under consideration must retain positive measure, orientation, adjacency, or causal accessibility. In a constructive setting, the lower bound of refinement is not literal zero, but the relevant dimensional primitive.

Thus, for a coordinate or refinement parameter ϵ, the constructive interpretation is not that the modeled object literally terminates at

$$\epsilon = 0, \tag{14}$$

but that refinement remains bounded below by the primitive completion P_n required for the modeled n-dimensional relation:

$$\epsilon \downarrow P_n. \tag{15}$$

Here P_n denotes the primitive completion required for nonzero n-dimensional measure.

More explicitly,

$$P_0 = \text{point}, \qquad P_1 = \text{straight segment}, \qquad P_2 = \text{triangle}, \qquad P_3 = \text{tetrahedron}. \tag{16}$$

Under balance symmetry,

$$P_2 = \text{equilateral triangle}, \qquad P_3 = \text{regular tetrahedron}. \tag{17}$$

This distinction is especially important for curves. A curve is not primitive; it is a derived object formed by refining locally straight relations. Therefore, a curve cannot be used to justify a limit operation that destroys the primitive structure required to construct the curve in the first place. Any curve admitted into a constructive model must be reconstructed from primitive-bounded segments or simplex relations.

> A constructive limit must terminate at the smallest admissible primitive that preserves the measure and relation being modeled. It may not terminate at zero if zero destroys the structure required for the modeled object to exist.

9 Minimal Constructive Closure in Two Dimensions

Definition 9.1 (Constructive two-dimensional primitive)**.** *A constructive two-dimensional primitive is a finite vertex-defined closed object with positive two-dimensional measure and the minimum number of boundary-defining vertices required to enclose area.*

Lemma 9.2. *Fewer than three vertices cannot enclose positive area.*

Proof. One point has no extension and zero area. Two distinct points define at most a line segment, which has zero two-dimensional measure. Hence fewer than three vertices cannot enclose positive area. □

Lemma 9.3. *Three non-collinear vertices define a minimal closed positive-area primitive.*

Proof. Let A, B, and C be three non-collinear points. Their convex hull is a triangle. Since the points are non-collinear, the vectors $B - A$ and $C - A$ are linearly independent, so the enclosed oriented area is nonzero. In a coordinate report this may be written as

$$A_{\triangle} = \frac{1}{2} \left| (B - A) \wedge (C - A) \right| > 0, \tag{18}$$

or, after selecting a base and height, as $A_{\triangle} = \frac{1}{2} bh > 0$. The factor $1/2$ is a coordinate report relative to a parallelogram or rectangular grammar; it does not make the triangle half of a primitive area unit. It shows that the triangle is the first finite closed area-consuming cell. □

Theorem 9.4 (Minimal constructive primitive of area)**.** *If positive area must be constructed from a nonzero finite vertex-defined closed primitive, then the minimal such primitive is a triangle.*

Proof. Fewer than three vertices cannot enclose positive area. Three non-collinear vertices can enclose positive area. Therefore, the triangle is the minimal finite area-consuming primitive. □

10 Balance and the Equilateral Triangle

Definition 10.1 (Balanced two-dimensional primitive)**.** *A balanced two-dimensional primitive is a minimal closed two-dimensional primitive whose boundary elements are equivalent under symmetry, so that no edge or vertex is privileged.*

Theorem 10.2 (Balanced minimal constructive primitive of area)**.** *The equilateral triangle is the balanced minimal constructive primitive of area.*

Proof. By the previous theorem, the minimal closed area-consuming primitive is a triangle. To impose balance, no side may be privileged over another, so all side lengths must be equal:

$$a = b = c. \tag{19}$$

A triangle with all side lengths equal is equilateral and has all angles equal to 60°. Therefore, the equilateral triangle is the balanced minimal constructive primitive of area. □

11 Minimal Constructive Closure in Three Dimensions

Definition 11.1 (Constructive three-dimensional primitive)**.** *A constructive three-dimensional primitive is a finite vertex-defined closed object with positive three-dimensional measure and the minimum number of boundary-defining vertices required to enclose volume.*

Lemma 11.2. *Fewer than four non-coplanar vertices cannot enclose positive volume.*

Proof. One point has no volume. Two points define at most a line. Three points define at most a plane or triangular face. Any set determined by fewer than four vertices lies in a lower-dimensional affine subspace and therefore has zero three-dimensional measure. Hence fewer than four non-coplanar vertices cannot enclose positive volume. □

Lemma 11.3. *Four non-coplanar vertices define a minimal closed positive-volume primitive.*

Proof. Let A, B, C, and D be four non-coplanar points. Their convex hull is a tetrahedron. Since D does not lie in the plane determined by A, B, and C, the vectors $B - A$, $C - A$, and $D - A$ are linearly independent. In a coordinate report, the enclosed volume may be written as

$$V_{\text{tet}} = \frac{1}{6} \left| \det(B - A,\, C - A,\, D - A) \right| > 0, \tag{20}$$

or, after selecting a triangular base and height, as $V_{\text{tet}} = \frac{1}{3} A_{ABC} h > 0$. Therefore, four non-coplanar vertices are sufficient to enclose positive volume. □

Theorem 11.4 (Minimal constructive primitive of volume)**.** *If positive volume must be constructed from a nonzero finite vertex-defined closed primitive, then the minimal such primitive is a tetrahedron.*

Proof. Fewer than four non-coplanar vertices cannot enclose positive volume. Four non-coplanar vertices can enclose positive volume. Therefore, the tetrahedron is the minimal finite volume-consuming primitive. □

12 Balance and the Regular Tetrahedron

Definition 12.1 (Balanced three-dimensional primitive)**.** *A balanced three-dimensional primitive is a minimal closed three-dimensional primitive whose edges, faces, and vertices are equivalent under symmetry, so that no edge, face, or vertex is privileged.*

Theorem 12.2 (Balanced minimal constructive primitive of volume)**.** *The regular tetrahedron is the balanced minimal constructive primitive of volume.*

Proof. By the previous theorem, the minimal closed volume-consuming primitive is a tetrahedron. To impose maximal balance, no edge or face may be privileged over another. Therefore, all six edge lengths must be equal. A tetrahedron with all six edge lengths equal has four congruent equilateral triangular faces and is regular. Therefore, the regular tetrahedron is the balanced minimal constructive primitive of volume. □

13 General Form: The Simplex Theorem

Definition 13.1 (n-simplex)**.** *An n-simplex is the convex hull of $n+1$ affinely independent points. Examples include the line segment in one dimension, the triangle in two dimensions, and the tetrahedron in three dimensions.*

Theorem 13.2 (Primitive Measure Theorem)**.** *Let an n-dimensional constructive primitive be defined as a finite vertex-defined closed object with positive n-dimensional measure and with the minimum number of vertices required to enclose such measure. Then the primitive is an n-simplex. Under maximal balance symmetry, the primitive is the regular n-simplex.*

Proof. A finite vertex-defined primitive with positive n-dimensional measure must contain at least $n+1$ affinely independent vertices; otherwise its vertices lie in an affine subspace of dimension less than n, and the bounded region has zero n-dimensional measure. The convex hull of $n+1$ affinely independent points is an n-simplex and has positive n-dimensional measure. Therefore, the minimal finite closed positive-measure primitive in dimension n is the n-simplex.

If maximal balance is imposed, the vertices and edges must be equivalent under the symmetry group of the primitive. This yields the regular n-simplex. Therefore, the balanced constructive primitives are the equilateral triangle in two dimensions and the regular tetrahedron in three dimensions. □

14 Cartesian Squares and Cubes as Composite Coordinate Units

A square is not the minimal closed two-dimensional primitive. It requires four vertices and can be decomposed into two triangles. Therefore, as a constructive primitive, it is composite relative to the triangle.

A cube is not the minimal closed three-dimensional primitive. It requires eight vertices and can be decomposed into tetrahedra. Therefore, as a constructive primitive, it is composite relative to the tetrahedron.

The common expressions

$$A = x^2, \qquad V = x^3 \tag{21}$$

are valid dimensional scaling relations inside an imposed coordinate grammar. They do not establish that squares and cubes are the primitive constructors of area and volume.

15 Accessible Measure Versus Coordinate Measure

A coordinate system may assign measure to a region without proving that the entire assigned measure is constructively accessible under the primitive geometry. This distinction is essential.

Cartesian coordinates imply that area and volume can be swept out by orthogonal displacement along independent axes. In such a representation, the coordinate plane is treated as if it were the natural carrier of area, and the coordinate cube is treated as if it were the natural carrier of volume. However, this is a property of the coordinate grammar, not a proof of primitive accessibility.

If the constructive primitive is simplex-based, then not every coordinate-accessible direction is necessarily primitive-accessible. Construction must proceed through the permitted relations of the primitive: vertices, edges, faces, orientation, closure, and adjacency. Thus, area and volume are not merely quantities assigned to regions; they are measures of reachable or constructible relation under the rules of the primitive geometry.

This is one reason bivectors are more appropriate than orthogonal planes for a primitive constructive account. An orthogonal plane is a coordinate abstraction. It presupposes rectangular decomposition. A bivector represents an oriented area element generated by two directed relations and is therefore closer to the primitive requirement of bounded, oriented relational construction.

In a simplex-based geometry, the primitive area carrier is not an infinite orthogonal plane but an oriented triangular relation. Likewise, the primitive volume carrier is not an orthogonal cube but a bounded tetrahedral relation. Coordinate area and coordinate volume may remain valid projections, but they do not establish that all projected area or volume is primitively accessible.

> Coordinate measure describes what an imposed measurement system can assign. Constructive measure describes what the primitive geometry can actually enclose, access, or propagate through.

16 Anti-Circularity of Limit Primitives

A limit cannot be used as the primitive justification for another limit without circularity. A limiting process is a formal operation performed over objects whose constructive meaning must already be defined. Therefore, the limiting operation cannot itself supply the primitive object whose structure it is being used to erase.

> Limits may refine a prior construction, but they may not serve as the primitive definition of the construction being refined.

Thus, a curve cannot be defined primitively by appeal to an infinitely smooth curve, a fluid cannot be defined primitively by appeal to an infinitely smooth continuum, and a spatial primitive cannot be defined by appeal to a zero-measure differential whose constructive contribution vanishes when treated literally.

The correct order is therefore:

1. define the primitive constructive relation;
2. determine the admissible finite refinement;
3. apply limits only as controlled abstractions over that construction;
4. preserve an assumption ledger identifying what the limit has suppressed.

Whenever this order is reversed, the model risks defining limits by limits. That may produce formal elegance, but it does not provide a primitive constructive account.

17 Relation to Calculus and Measure Theory

Calculus does not obtain positive area by first setting every width to zero and then summing. In a Riemann sum, each finite-stage subdivision has nonzero width:

$$\sum_{i=1}^{m} f(x_i^*)\Delta x_i. \tag{22}$$

Only after finite nonzero contributions are summed is the limit taken:

$$\int_a^b f(x)\,dx = \lim_{\max \Delta x_i \to 0} \sum_{i=1}^{m} f(x_i^*)\Delta x_i. \tag{23}$$

Thus calculus does not demonstrate that literal zero-width primitives construct positive measure. It demonstrates that sums of nonzero finite subdivisions may converge to a stable limiting value.

Measure theory can assign positive measure to an interval even though each individual point has measure zero. This does not contradict the zero-contribution lemmas, because measure theory is countably additive, not naively uncountably additive over individual point measures. The continuum is treated as an axiomatically completed object, not as a constructive finite or countable sum of zero-measure parts.

Saying that a continuum may be represented as an uncountable set of points is not the same as saying positive length, area, or volume is constructively produced by adding zero-measure primitives.

18 Circles, Spheres, and Smooth Ideals

Under classical isoperimetric analysis, circles and spheres optimize area or volume relative to boundary. However, these are smooth continuum ideals. They are not minimal finite closed primitives. The present theorem concerns minimal constructive closure, not smooth optimization in an already-assumed continuum.

A circle may be approached as a limiting process over polygons, but it is not the first closed area-consuming polygon. A sphere may be approached by smooth or polyhedral approximation, but it is not the first closed volume-consuming polyhedron. Minimal constructive closure belongs to the simplex.

19 Implications for Geometric and Physical Modeling

If a model treats points, lines, or zero-thickness planes as primitive, then any behavior requiring boundary, orientation, resistance, adjacency, chirality, or internal relation must be introduced later as abstract structure. A constructive simplex-based model begins with finite relational geometry:

- vertices,
- edges,
- faces,
- closure,
- orientation,
- adjacency,
- nonzero measure.

This does not by itself prove a complete physical theory. It establishes that if primitives are required to consume space constructively, then point-based or Cartesian-unit-based ontology is not primitive. It is a projection or coordinate convenience layered over a deeper simplex construction.

20 Conclusion

A zero-measure primitive cannot construct positive measure by ordinary finite or countable aggregation. Therefore, if area and volume are interpreted constructively rather than merely axiomatically or coordinately, their primitives must already possess nonzero measure and dimensional closure.

The minimum finite closed positive-measure primitive in two dimensions is the triangle. Under balance symmetry, it is the equilateral triangle. The minimum finite closed positive-measure primitive in three dimensions is the tetrahedron. Under balance symmetry, it is the regular tetrahedron.

Therefore, the primitive constructive structure of consumed space is simplex-based, not Cartesian. Cartesian coordinates remain powerful tools for measurement and calculation, but they should not be mistaken for the primitive geometry of consumed space.

Core Thesis

Positive area cannot be constructively built from zero-area primitives, and positive volume cannot be constructively built from zero-volume primitives. Therefore, if the primitive is nonzero and space-consuming, the minimal finite constructive primitives are the equilateral triangle in two dimensions and the regular tetrahedron in three dimensions.

Statements and Declarations

Funding

No funding was received for conducting this study or preparing this manuscript.

Competing Interests

The author declares no relevant financial or non-financial interests.

Data Availability

No datasets were generated or analyzed during the current study.

Author Contribution

Joseph Fredrickson is the sole author and was responsible for the conception, development, formal argument, writing, and revision of the manuscript.

AI Assistance Statement

AI-assisted tools were used for manuscript preparation support, including organization, copy editing, formatting assistance, terminology refinement, and alignment with relevant scholarly conventions. The theory, core arguments, definitions, theorem structure, interpretation, and any computational testing were developed by the author. The author reviewed and revised the manuscript and accepts full responsibility for the final content.

Biography

Joseph Fredrickson is an independent researcher based in Orem, Utah, USA. He holds a Bachelor of Science degree in Mechanical Engineering from Brigham Young University and has professional experience in engineering analysis, computational modeling, search engine optimization, artificial intelligence workflows, and technical writing. His current research focuses on constructive geometry, primitive measure, causal modeling, and relation-preserving computation. His work investigates the distinction between coordinate representation and constructive primitives, with applications to geometry, calculus, number theory, physics, and computational efficiency. He has developed computational prototypes that preserve structured relations rather than expanding them into coordinate-heavy representations, including constructive runtime tools for verified fast-path computation. His broader research program examines whether mathematical and physical models become more explanatory and computationally efficient when primitive closure, finite relation, and explicit extension costs are treated as first-class objects. He is the author of the manuscript "The Primitive Measure Theorem," which formalizes a constructive distinction between coordinate measure and minimal simplex closure. ORCID: 0009-0005-3327-4408.

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